

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 12-2025: A 56-Year-Old Woman with Sore Throat and Rash

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PRESENTATION OF CASE

Dr. Sarah E. Kler (Medicine): A 56-year-old woman was evaluated in the emergency department of this hospital because of sore throat and rash.

The patient had been in her usual state of health until 2 days before the current presentation, when sore throat and pruritus on the chest developed. There was a history of depression with psychosis, and she resided in a homeless shelter. Despite pruritus, the patient did not inspect the skin on her chest.

On the day of the current presentation, while the patient was looking in the mirror in a public bathroom, she noticed a rash on her face, which prompted her to look at the rest of her skin. She saw that the rash also involved the chest, abdomen, and legs. She presented to the emergency department of this hospital for evaluation.

In the emergency department, the patient reported ongoing pruritus and sore throat. She noted that several residents at the homeless shelter had appeared to be scratching themselves in the week before the current presentation, and she had heard an employee at the shelter discussing lice and scabies with another shelter resident. There was no known exposure to new soaps, detergents, or lotions. She did not sleep outside or spend time in the woods. She was not sexually active. She had no fever, malaise, headache, confusion, changes in strength or sensation, dyspnea, or cough. In addition to her history of depression with psychosis, she had a history of dyslipidemia and metabolic dysfunction–associated steatohepatitis. Medications included risperidone and benztropine. There were no known adverse reactions to medications. She did not smoke tobacco, drink alcohol, or use illicit drugs.

The temporal temperature was 38.4°C, the blood pressure 134/72 mm Hg, the pulse 83 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 96% while the patient was breathing ambient air. She was alert and cooperative. Diffuse pink papules were present on the forehead, nose, cheeks, and chin (Fig. 1A). An ulcer was present on the anterior upper gum (Fig. 1B). Excoriated papules and vesicles were seen on the back, chest, and abdomen, with involvement of the inframammary and inguinal folds (Fig. 1C through 1F). Pink

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CME



papules were noted on the dorsal aspect of the hands and the palms (not shown). The lung sounds were normal.

The blood levels of electrolytes and alkaline phosphatase were normal, as were the results of kidney-function tests. The aspartate aminotransferase level was 44 U per liter (reference range, 9 to 32), the alanine aminotransferase level 60 U per liter (reference range, 7 to 33), and the albumin level 3.2 g per deciliter (reference range, 3.3 to 5.0). The white-cell count was 5300 cells per microliter with 5.3% atypical lymphocytes; the remainder of the complete blood count was normal.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. John Trinidad: This 56-year-old woman with depression with psychosis presented to this hospital with a 2-day history of pruritic rash and sore throat. She was residing in a homeless shelter and reported that other residents might have had similar symptoms. An examination revealed scattered erythematous macules, papules, excoriated papules, and few vesicles with erythematous bases on the face, chest, back, and abdomen. Shallow ulceration was seen on the oral mucosa, a finding consistent with enanthem. In constructing a differential diagnosis for this patient's rash, I will begin by considering how



Figure 1. Clinical Photographs.

Diffuse pink papules are present on the forehead, nose, cheeks, and chin (Panel A). An ulcer is present on the anterior upper gum (Panel B). Excoriated papules and vesicles are shown on the back (Panel C), chest (Panel D), and abdomen (Panel E), with involvement of the inframammary and inguinal folds (Panel F). The patient's hands are shown in Panel B and at the top of Panel F.

her living conditions affect her risk of various dermatologic conditions.

HOMELESSNESS AND DERMATOLOGIC DISEASE

Persons experiencing homelessness are disproportionately affected by dermatologic conditions, possibly owing to limited access to hygiene facilities, exposure to communal living environments, and barriers to health care access.¹ The use of shared spaces and bedding increases the risk of parasitic infestations. In persons experiencing homelessness, chronic conditions such as eczema, psoriasis, and seborrheic dermatitis are often severe and untreated. Substance use, particularly injection-drug use, can increase susceptibility to skin infections. Communicable diseases such as respiratory, gastrointestinal, and sexually transmitted infections are more common and more quickly disseminated among persons experiencing homelessness, especially those in communal living environments, than in the general population. Parasitic infestations are a particularly important consideration in this patient, given her residence in a shelter and her presentation with diffuse rash and pruritus.

PARASITIC INFESTATIONS

Scabies

Scabies is a parasitic infestation caused by the mite *Sarcoptes scabiei* var. *hominis*, which burrows into the skin to lay eggs. Subtypes include classic, nodular, and crusted (Norwegian) scabies. Patients with classic scabies present with burrows, erythematous papules, and excoriations that typically involve the interdigital web spaces, wrists, ankles, axillae, genitals, and periumbilical regions. Patients with crusted scabies present with hyperkeratotic plaques with a “sand on skin” morphologic appearance that often involve the hands and feet; this subtype is more common in immunosuppressed persons, especially those with advanced acquired immunodeficiency syndrome.

Scabies affects persons at all socioeconomic levels, with the estimated prevalence ranging from 0.18% to as high as 76.9% across studies.² The prevalence is higher in countries with a low human development index, as well as in populations displaced by conflict or migration. The diagnosis of scabies is typically based on the presence of rash with the characteristic morphologic appearance and distribution, along with

pruritus, which is often worse at night. A skin scraping with mineral oil or a potassium hydroxide preparation can be performed to detect the presence of mites or their eggs or feces, but the sensitivity of this bedside test depends on the experience of the examiner. Direct in vivo examination with dermoscopy can be useful but requires training. Findings on dermoscopy include the delta-wing sign (a feature indicative of the mite body) and the double-edged sign (a feature outlining the burrow and caused by feces). The mites are very small — for example, they are smaller than dermatoglyphs (fingerprint lines) and eccrine ducts (sweat ducts) — and may be easily overlooked.

This patient's rash involved the face, which is not consistent with the typical scabies rash distribution. Severe scabies reactions can cause widespread eruptions similar to that seen in this patient, but vesicles would be an uncommon manifestation.

Crab Lice

The crab louse, *Phthirus pubis*, is an obligate parasite that attaches to hair shafts, primarily in the pubic region, axillae, eyelashes, and beard. Findings associated with crab lice infestation include pruritus, erythematous papules, nits (eggs) on hair shafts, and rarely, maculae ceruleae (blue-gray macules caused by bilirubin breakdown). The incidence of crab lice infestation ranges from 0.3 to 4.6% of the population worldwide, with a prevalence of approximately 2% in adults.³ The incidence may be increased in populations affected by war, disasters, or overcrowding. The clinical identification of lice or nits is sufficient for diagnosis. The lice resemble crabs; they have broad bodies and often hang by one side on hair shafts. Crab lice infestation would not cause a generalized rash or vesicles, so this diagnosis is unlikely in this patient.

Body Lice

The body louse, *Pediculus humanus corporis*, is a parasite that typically resides in clothing seams and moves to the skin to feed. Patients with body lice infestation present with diffuse pruritic papules and excoriations; secondary bacterial infections can develop. Body lice infestation — unlike scabies and infestations of crab lice and head lice — is strongly associated with homelessness, with the prevalence ranging from 19.1 to 68.0%

among persons experiencing homelessness.⁴ The diagnosis is often confirmed by direct examination of clothing seams for live lice or nits. In severe cases, the lice can be directly visualized on the patient's skin and bedding. The skin manifestations are often nonspecific and are due to intense pruritus. The diffuse papules seen in this patient could fit with the diagnosis of body lice infestation, but the vesicular component and facial involvement of the rash are atypical of this diagnosis.

AVOIDING ANCHORING BIAS

Although parasitic infestations are prevalent among persons experiencing homelessness, there is a risk of anchoring on the patient's residence in a homeless shelter where other residents might have had pruritus. Anchoring bias — the cognitive tendency to lock onto salient features of a patient's initial presentation without adjusting after additional information is obtained — can hinder diagnostic accuracy.⁵ In my opinion, the effects of anchoring bias can be exacerbated in marginalized populations and thus may widen disparities in health outcomes.

In this patient, anchoring on the possibility of parasitic infestation might delay consideration and recognition of other causes of rash, such as viral or inflammatory conditions. A delay in diagnosis could increase the risk of the disease spreading if the rash is due to an infection. Recognizing the potential for anchoring bias in the evaluation of persons experiencing homelessness is a critical first step for physicians working to use cognitive strategies to avoid anchoring. In this case, none of the previously described parasitic infestations can fully account for the patient's clinical presentation. At this point, I will focus on the morphologic appearance and distribution of the skin lesions to refine the differential diagnosis.

DRUG REACTIONS

A drug reaction should be considered in any patient with a diffuse rash. Many drug-related eruptions can be classified as severe cutaneous adverse reactions to medication, including life-threatening conditions such as the Stevens–Johnson syndrome, toxic epidermal necrolysis, and drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome. Early identifica-

tion of the reaction, discontinuation of the medication, and treatment with immunosuppressive agents are essential steps for decreasing the risk of complications or death associated with these conditions.

The patient had been taking risperidone and benztropine, drugs that are unlikely to cause dermatologic disease. A drug reaction is also unlikely in this patient because her rash included macules and papules that were distinct and did not coalesce into patches or plaques. In addition, the vesicles and enanthem would be unusual features of a drug reaction and are more suggestive of certain viral infections.

VIRAL INFECTIONS

Measles

Patients with measles present with a morbilliform rash consisting of macules and papules that coalesce into patches and plaques. The rash typically begins on the face and spreads in a cephalocaudal manner. Koplik spots (bluish-white papules on the buccal mucosa) are a hallmark feature of measles. Systemic symptoms include a high temperature, cough, coryza, and conjunctivitis.

The incidence of measles is low but increasing, and outbreaks have occurred because of a decrease in vaccination.⁶ An estimated 10.3 million persons globally were infected with measles in 2023.⁷ A measles outbreak in West Texas that began in early 2025 has expanded to 327 cases across 15 counties, with 40 hospitalizations and one death (as of March 25).⁸

Measles is unlikely in this patient because her rash had macules and papules that were distinct and did not coalesce. Her oral enanthem was a shallow erosion and was not bluish-white. In addition, the presence of vesicles is not consistent with measles.

Rubella

Like measles, rubella is more common in unvaccinated persons. Rubella resembles measles in terms of the rash, but the systemic symptoms are milder. Postauricular and occipital lymphadenopathy is a distinguishing feature of rubella. This patient's rash, with macules and papules that did not coalesce, was not consistent with rubella, nor was the presence of vesicles and enanthem.

Acute HIV Infection

Acute retroviral syndrome develops in up to 80% of patients with a new human immunodeficiency virus (HIV) infection,⁹ typically occurring 2 to 6 weeks after exposure to the virus. Symptoms last days to weeks and include fever, myalgia, pharyngitis, lymphadenopathy, and night sweats. In 40 to 80% of cases, a morbilliform rash develops that is most prominent on the face and trunk and lasts 4 to 5 days. Oral and genital ulcers may be present. This patient had one oral ulcer but did not have a true morbilliform eruption, which makes primary HIV infection unlikely.

Herpes Simplex Virus Infection

Patients with herpes simplex virus (HSV) infection typically present with localized vesicular lesions that progress to pustules and crusts. Oral or genital mucosal involvement is common, and the morphologic appearance can vary depending on the HSV subtype and means of acquisition. Although vesicular lesions were present in this patient, the diffuse nature of her rash would be atypical of both primary HSV infection and HSV reactivation. Disseminated HSV infection can occur in immunocompromised persons, causing a systemic illness that is characterized by fever, headache, neurologic findings, and other end-organ disease. The skin manifestations associated with disseminated HSV infection are vesicular lesions in a widespread distribution that can progress to bullae and large ulcers. The absence of known immunocompromise in this patient makes disseminated HSV infection unlikely.

Mpox

Patients with mpox typically present with a localized vesiculopustular rash with a stereotyped progression.¹⁰ Established lesions are often deep-seated, well-circumscribed, and umbilicated. The rash can affect the face, hands, feet, chest, genitals, and perianal region. Although the rash is usually localized, it can be diffuse in some cases. In the 2022 outbreak, many patients presented with atypical features, including fewer lesions, proctitis, and minimal systemic symptoms. The incubation period ranges from 3 to 17 days, and affected patients remain contagious until all the lesions have scabbed and healed. In this patient, the morphologic appearance of the

rash was not consistent with a typical case of mpox; the lesions were diffuse and did not have the characteristic deep-seated, well-circumscribed, umbilicated appearance.

Varicella–Zoster Virus Infection

Varicella–zoster virus (VZV) infection is a highly contagious herpesvirus infection that causes two distinct clinical syndromes: varicella (chickenpox) and herpes zoster (shingles).⁹ Varicella is caused by primary VZV infection, whereas herpes zoster results from VZV reactivation. After primary infection occurs, VZV establishes latency in the dorsal-root ganglia and can reactivate later in life.

Primary VZV infection typically affects children. Widespread childhood vaccination has reduced both the incidence and the severity of disease, given that breakthrough cases in vaccinated persons tend to be milder. The rash associated with varicella is polymorphic, including erythematous macules, papules, vesicles, and crusts at various stages at the same time. Lesions typically spread in a cephalocaudal manner, starting on the scalp and face and progressing to the trunk, arms, and legs. Additional symptoms include fever and malaise. The rash is accompanied by pruritus, as opposed to pain.

VZV reactivation typically occurs in adults, with increasing age being the most important risk factor. The rash associated with herpes zoster develops in a dermatomal distribution, starting with painful vesicular lesions that proceed through numerous stages but appear and evolve synchronously within the dermatome affected. In the context of disseminated disease, which can be seen in immunocompromised persons, the rash can be widespread.

This patient did not have known risk factors for immunocompromise and did not describe the rash as painful, so a diagnosis of disseminated herpes zoster was unlikely. Varicella is rare in adults, but it can occur and is associated with severe disease and complications such as hepatitis and pneumonitis. Therefore, prompt recognition of the diagnosis and initiation of treatment is important. In this patient, the presence of vesicles, enanthem, and polymorphic lesions, as well as elevated hepatic-enzyme levels, is most consistent with primary VZV infection.

To establish the diagnosis, I would unroof and scrape a prominent vesicular lesion to obtain a

specimen for VZV-specific nucleic acid testing or direct fluorescent antibody testing. Given that disseminated HSV infection is also a possibility, I would send a specimen for HSV-specific nucleic acid testing or direct fluorescent antibody testing. A biopsy of the rash would be appropriate if pathological examination of the specimen could be performed within a short time frame, allowing for the rapid initiation of antiviral therapy and limiting the spread of disease.

DR. JOHN TRINIDAD'S DIAGNOSIS

Primary varicella–zoster virus infection.

DIAGNOSTIC TESTING

Dr. Ting Zhao: Examination of a punch-biopsy specimen of the skin on the right side of the chest revealed vesicular and acantholytic changes involving the epidermis and hair-follicle epithelium (Fig. 2A and 2B). There was marked epidermal necrosis; keratinocytes had prominent viral cytopathic effects, including multinucleation, nuclear molding, and chromatin margination (Fig. 2C and 2D). Immunohistochemical staining for VZV showed strong nuclear and cytoplasmic positivity (Fig. 2E). In contrast, immunohistochemical staining for HSV-1 and HSV-2 was negative (Fig. 2F). Nucleic acid testing for VZV DNA performed on the skin-biopsy specimen confirmed the presence of VZV, whereas nucleic acid testing for HSV-1 and HSV-2 DNA was negative. Bacterial culture of the biopsy specimen grew *Cutibacterium acnes* (formerly *Propionibacterium acnes*), a component of normal skin flora that resides predominantly in sebaceous glands. Mycobacterial and fungal cultures were negative. Serologic testing showed no detectable VZV IgM or IgG, and screening for HIV-1 antigens and for HIV-1 and HIV-2 antibodies was also negative. Overall, these findings are diagnostic of VZV infection.

The negative testing for VZV IgM and IgG indicated that the patient had no previous exposure or immunity to the virus, suggesting a diagnosis of primary VZV infection. The absence of detectable IgM may reflect that the blood sample was obtained before the development of a measurable immune response. VZV IgM and IgG typically appear 2 to 5 days after the onset of the rash, with peak titers observed at 2 to 3 weeks. VZV IgM levels decline rapidly and are usually

undetectable within 1 year, whereas the IgG levels decrease more gradually, often remaining detectable for several years.¹¹

PATHOLOGICAL DIAGNOSIS

Primary varicella–zoster virus infection.

DISCUSSION OF MANAGEMENT

Dr. Kimon C. Zachary: Many of us have experienced chickenpox as an uncomfortable but relatively inconsequential rite of passage during childhood. However, we are obliged to remember that this highly transmissible infection can lead to serious complications.

Effective infection control follows three steps: identify, isolate, and inform. Given that primary VZV infection conveys a high level of lifelong immunity to a second episode of clinical disease, chickenpox has long been uncommon among adults in the United States. The incidence among children has plummeted with widespread uptake of the varicella vaccine, which after two doses is approximately 95% effective in preventing clinical disease, although not subclinical infection.¹² Many clinicians therefore may be relatively unfamiliar with the clinical presentation of chickenpox, and we must remember to consider it in any patient presenting with a compatible rash. In this patient, there was an approximately 24-hour delay in recognition. Although more than 95% of adults born in the United States before 1980 have had chickenpox and are protected from a second episode, some have not had primary VZV infection and have not been vaccinated against it.¹³ The occurrence of a second episode of chickenpox, albeit very rare, is well-described, even in ostensibly immunocompetent persons.

VZV is transmitted primarily through direct contact with skin lesions of affected patients or inhalation of aerosols from respiratory secretions.^{14,15} Aerosolization of the virus from skin lesions can also occur. Therefore, in addition to assiduous observation of standard precautions, implementation of contact and airborne isolation precautions is done as soon as possible.¹⁶ The patient is placed in a negative-pressure isolation room; if such a room is not immediately available, a private room is used, with the door kept closed, and the patient is asked to wear a surgical mask. Health care personnel, even those regarded as im-

immune to varicella, use a fit-tested respirator and wear a gown and gloves. Transmission can occur until all the skin lesions are dry and crusted, and these isolation precautions are observed until then.

Cases of chickenpox are reported to local infection-control and public health authorities. Nonimmune persons who have been exposed to the virus and who are not immunocompromised,

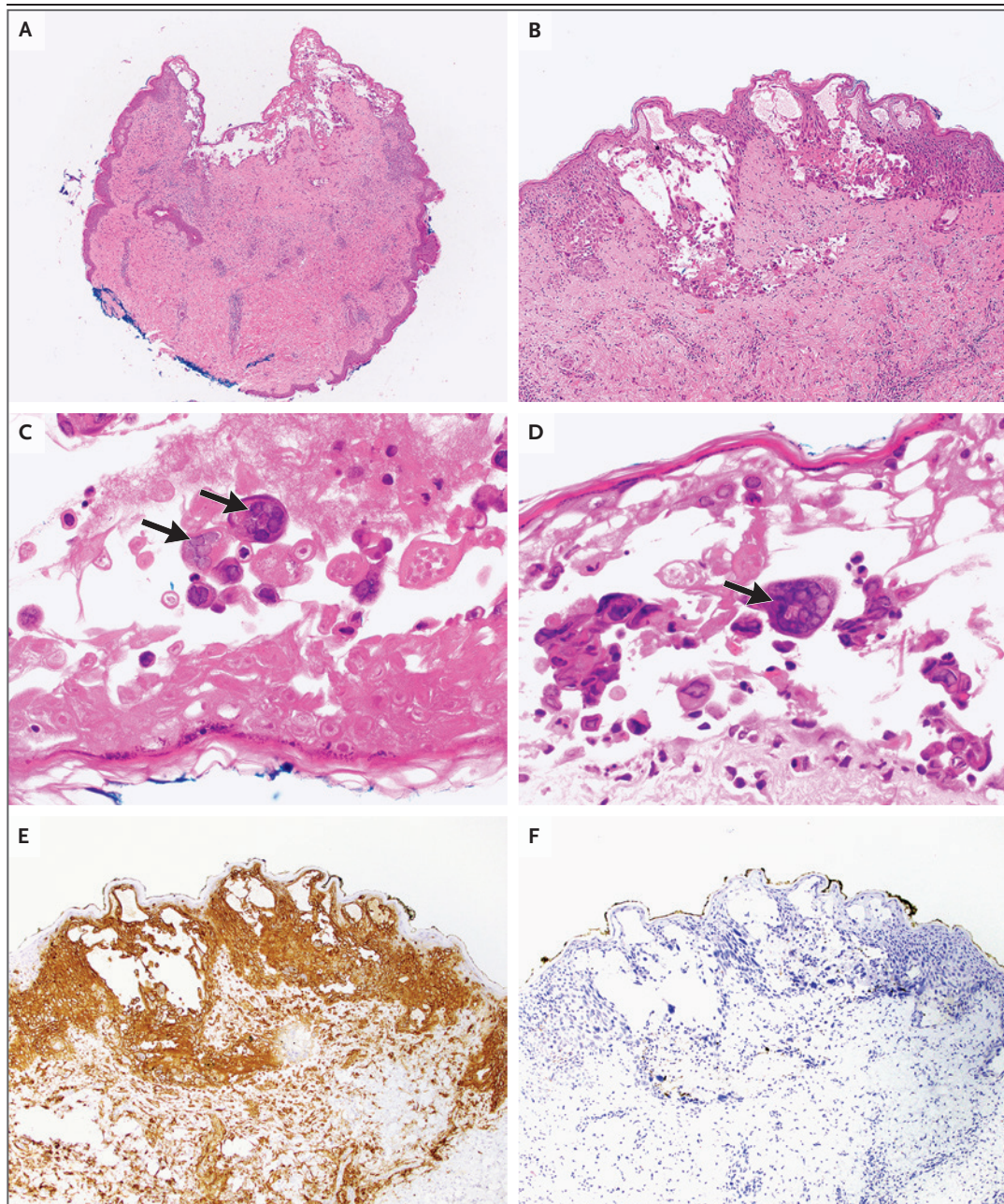


Figure 2. Biopsy Specimen of the Skin on the Right Side of the Chest.

Hematoxylin and eosin staining shows vesicular and acantholytic changes involving the epidermis and hair-follicle epithelium (Panels A and B). Keratinocytes have prominent viral cytopathic effects, including multinucleation, nuclear molding, and chromatin margination (Panels C and D, arrows). Immunohistochemical staining for varicella-zoster virus shows strong nuclear and cytoplasmic positivity (Panel E), whereas immunohistochemical staining for herpes simplex virus is negative (Panel F).

pregnant, or younger than 12 months of age receive postexposure prophylaxis with the vaccine, ideally within 5 days after exposure.¹⁷ Nonimmune exposed persons who are not candidates for the vaccine receive varicella–zoster immune globulin (VZIG) as soon as possible within 10 days after exposure.¹⁸ Antiviral prophylaxis is another option. Nonimmune health care personnel are furloughed from work from day 8 after exposure through day 21; the furlough is extended until day 28 if VZIG is administered.¹⁹

Except for healthy children 12 years of age or younger, patients with chickenpox receive antiviral therapy with intravenous acyclovir or oral valacyclovir, with the choice of agent dependent on the severity of the illness and the vulnerability of the host. Antiviral therapy is administered for reducing the severity of disease and the risk of complications, including pneumonitis and central nervous system involvement (meningitis, encephalitis, myelitis, or intracranial vasculopathy). Vigilance for bacterial superinfection of skin lesions, particularly superinfection with group A streptococcus, is necessary. Antiviral treatment is initiated within 24 hours after the onset of rash and is continued until all the skin lesions are dry and crusted, usually approximately 1 week. The benefit of treatment initiated at a later time point is uncertain; results from clinical trials have suggested no effect on the quantity or duration of skin lesions, although these studies were not powered to detect a difference in complications.^{20,21}

In this patient, treatment was started approximately 24 hours after presentation, which was more than 48 hours after the onset of rash, and was continued for 17 days. Given the lack of an alternative place in which to isolate while contagious, the patient remained hospitalized for the duration of her treatment and for 6 days afterward, out of concern that her skin lesions were not fully dry and crusted. This course was more prolonged than expected; it is possible that ex-

coriated crusts were mistaken for active lesions. There were no complications during the hospitalization. It should be noted that varicella-induced intracranial vasculopathy may manifest with a stroke-like syndrome several weeks after the acute illness.

Dr. Kler: One month after discharge, the patient established care at another hospital in Boston. The skin lesions were noted to have faded, and the results of a neurologic examination were normal. Since then, she has been hospitalized multiple times for psychiatric illness.

CASE RECORDS EDITORS' NOTE — LESSONS LEARNED

1. Persons experiencing homelessness are disproportionately affected by dermatologic conditions. Although parasitic infestations are prevalent in this population, anchoring on infestations as the cause of diffuse rash can delay the diagnosis of viral infection.

2. Primary VZV infection is uncommon in adults but should be considered in immunocompetent patients who have a diffuse rash that is compatible with varicella.

FINAL DIAGNOSIS

Primary varicella–zoster virus infection.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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